Nama: Hendry Hermawan

NIM: 2410511037

Laporan UTS PPOL

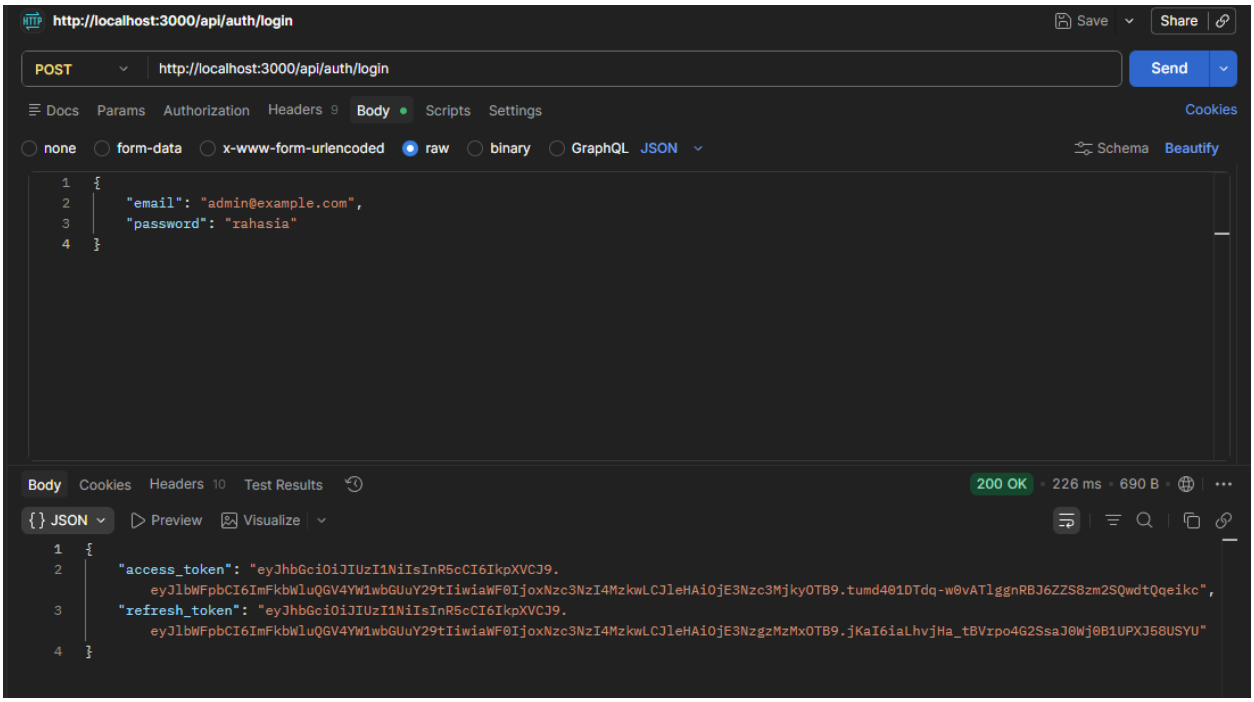
Auth Service: Dipisah agar keamanan tersentralisasi. Jika ingin ganti metode login (misal ke OAuth), hanya satu servis yang diubah.

Employee Service (Laravel): Data pegawai bersifat statis dan butuh validasi relasional yang kuat (maka pakai framework PHP yang matang).

Attendance Service (Node.js): Data absensi memiliki trafik tinggi (setiap pegawai absen barengan di pagi hari), Node.js lebih efisien menangani banyak koneksi ringan secara bersamaan daripada PHP.

Tes Postman:

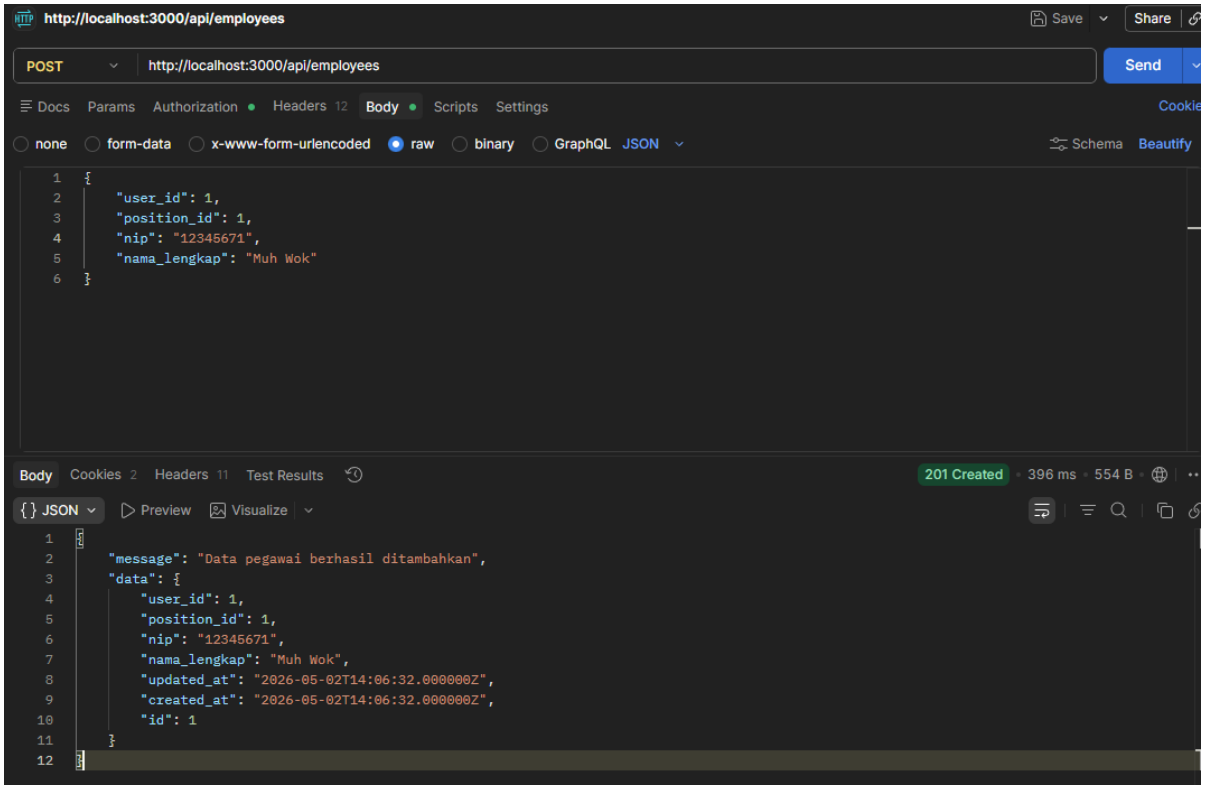
1. Auth Service: Login
   Method: POST
   URL: http://localhost:3000/api/auth/login
   Target: Status 200 OK dan muncul token.



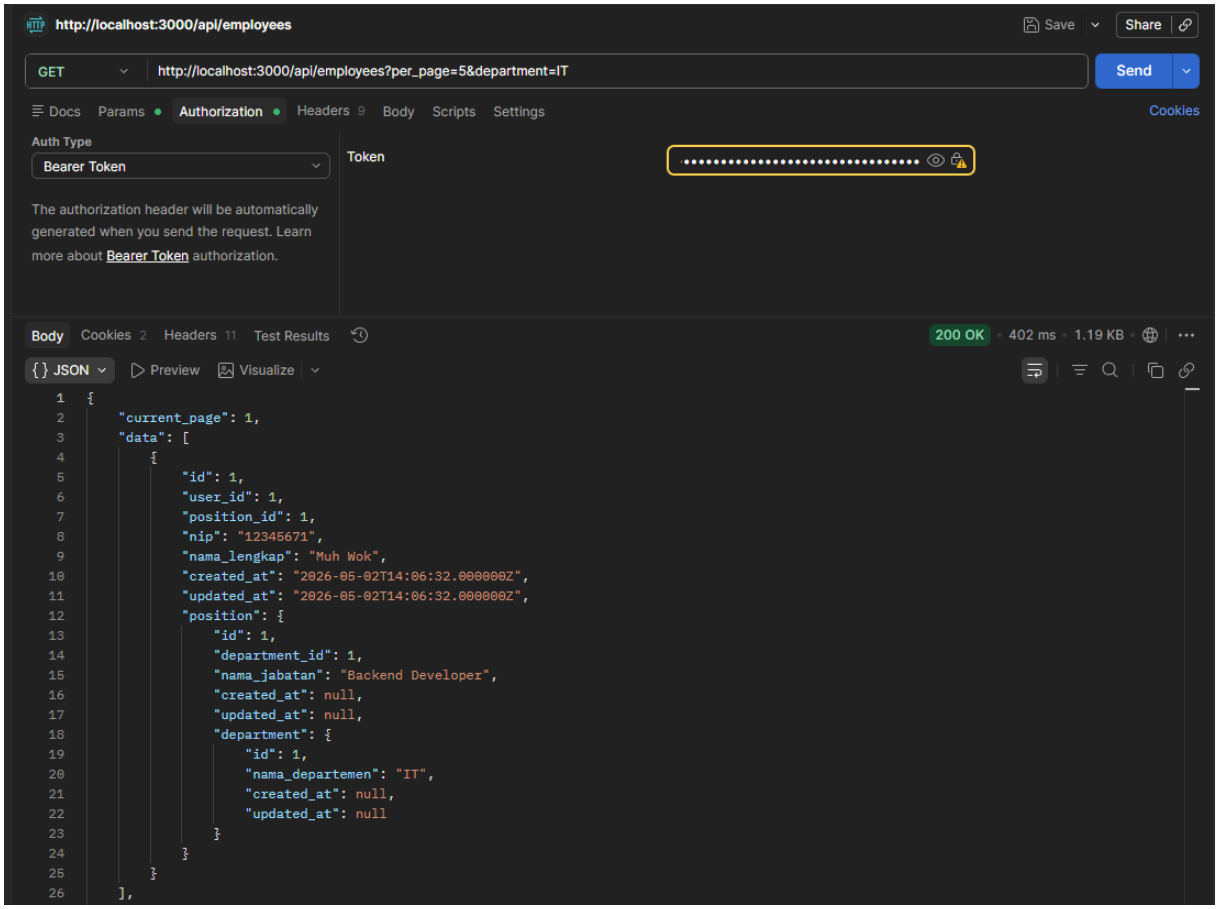
2. Employee Service: Create Employee

   Menambahkan data pegawai baru ke database MySQL.

   - Method: POST
   - URL: http://localhost:3000/api/employees
   - Auth: Type: Bearer Token
   - Target: Status 201 Created dan muncul data pegawai yang baru dibuat.



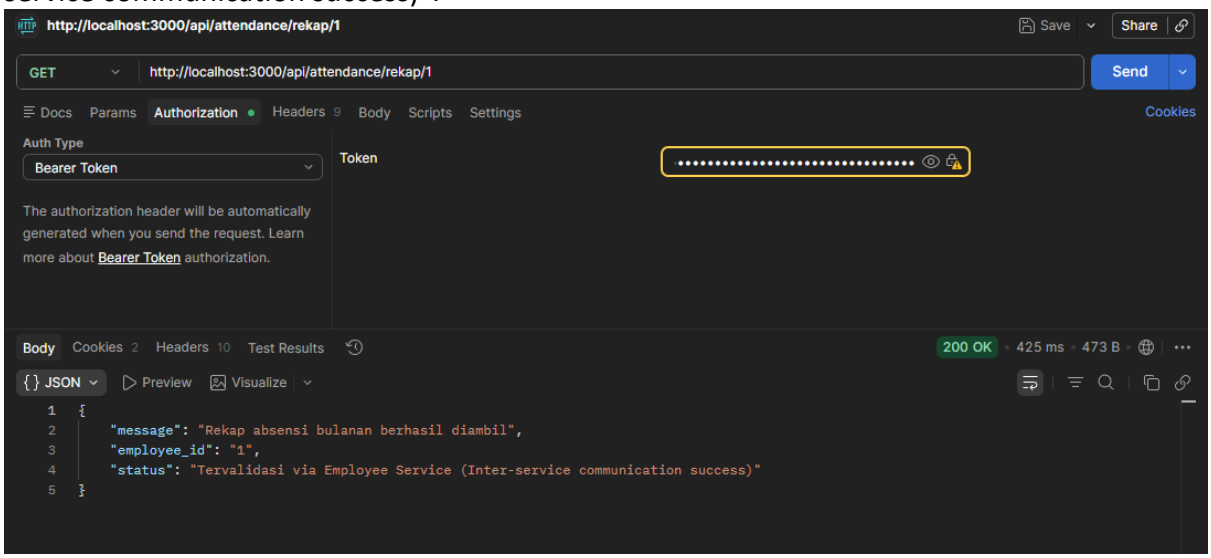
3. Employee Service: Get All (Paging & Filtering)
   Mengambil daftar pegawai dengan fitur filter departemen dan jumlah data per halaman (Syarat UTS).
   - Method: GET
   - URL: http://localhost:3000/api/employees?per_page=5&department=IT
   - Auth: Bearer Token (Masih aktif).
   - Target: Status 200 OK.



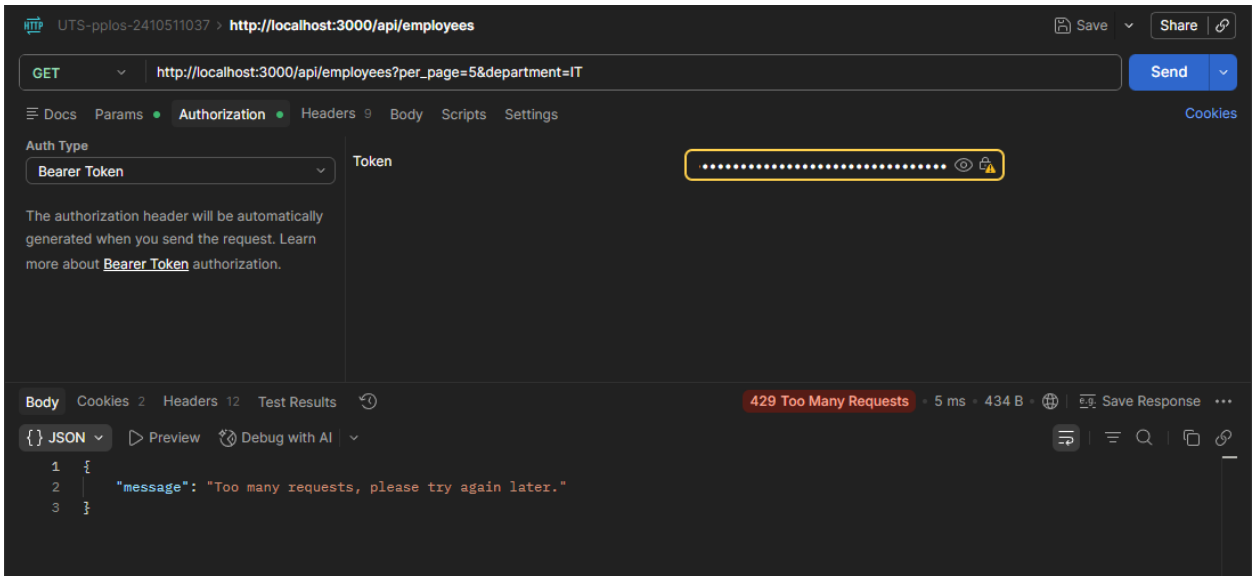
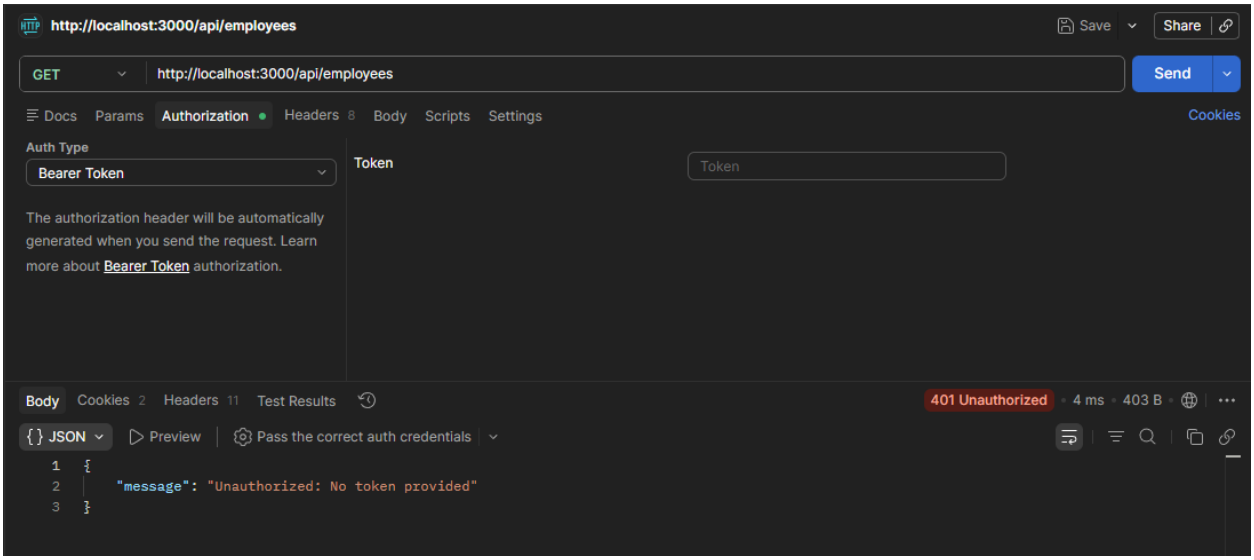
4. Attendance Service: Get Rekap (Inter-service)

   Mengecek apakah servis Node.js bisa memvalidasi data ke servis Laravel (Syarat UTS: Inter-service communication).

   - Method: GET
   - URL: http://localhost:3000/api/attendance/rekap/1
   - Auth: Bearer Token.
   - Target: Status 200 OK. Muncul pesan: "Tervalidasi via Employee Service (Inter-service communication success)".



5. Gateway: Rate Limiting Security
   Memastikan satu IP tidak bisa melakukan *spam* request berlebihan
   - Method: GET (Bebas pada endpoint mana saja).
   - URL: http://localhost:3000/api/employees
   - Cara Cek: Klik tombol Send berkali-kali dengan sangat cepat (sebanyak 60 kali dalam 1 menit).
   - Target: Pada klik ke-61, status akan berubah menjadi 429 Too Many Requests dengan pesan "Too many requests, please try again later."



6. Gateway: JWT Authorization Check
   Memastikan *endpoint* terlindungi dari akses tanpa izin.
   - Method: GET
   - URL: http://localhost:3000/api/employees
   - Auth: No Auth.
   - Target: Status 401 Unauthorized dengan pesan "No token provided".